

LEADER & ACHIEVER COURSE

PHASE : MLB, MLK, MLM, MLP, MLQ, MLR, MAZE, MAZF, MAZG, MAZH, MAZM, MAZP, MAZQ, MAZR, MAZU, MAAZ, MAAW, SPARK PRO

TARGET : PRE MEDICAL 2025

Test Type : **MAJOR**

Test Pattern : **NEET (UG)**

TEST DATE : 21-03-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	4	2	4	1	1	1	3	2	2	1	4	1	4	2	3	3	3	4	1	2	1	2	4	2	2	3	2	2	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	3	4	4	4	1	2	1	2	3	4	3	4	4	3	2	3	2	3	3	1	1	2	2	2	2	3	3	2	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	3	4	4	4	2	1	4	2	1	2	2	2	3	3	2	3	3	3	1	2	1	2	2	3	1	1	4	4	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	1	2	3	2	3	2	4	3	2	2	4	2	4	2	1	3	1	1	2	4	3	1	4	2	3	2	1	1	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	1	2	1	1	2	1	2	2	3	3	2	2	4	1	2	4	4	3	3	1	1	3	1	4	2	2	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	4	4	1	1	3	2	2	3	1	4	1	2	1	2	1	1	2	4	4	3	2	2	3	3	3	1	3	1	4

HINT - SHEET

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| <p>58. Ans (3)
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SOLUTION/EXPLANATION/CALCULATION:
The hormone that is synthesized elsewhere but stored and released by the master gland (pituitary gland) is Antidiuretic hormone (ADH).
ADH is synthesized in the hypothalamus but stored in the posterior pituitary gland. It is released when the body needs to conserve water, such as during dehydration or low blood pressure.</p> <p>73. Ans (2)
NCERT Pg.# 335</p> | <p>75. Ans (3)
NCERT Pg. # 219</p> <p>76. Ans (2)
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NCERT Pg. No. # 227</p> <p>90. Ans (2)
NCERT Pg. No. # 227</p> <p>91. Ans (2)
<div style="border: 1px solid black; padding: 5px; display: inline-block;">$KE = \frac{f}{2}nRT$</div></p> |
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92. Ans (1)

From FLOT

$$Q = \Delta U + W$$

For path 'iaf'

$$50 = \Delta U + 20$$

$$\Delta U = 30 \text{ Cal}$$

For path 'ibf'

$$\Delta U = 30 \text{ Cal.}$$

$$Q = \Delta U + W$$

$$W = 36 - 30$$

$$= 6 \text{ cal}$$

93. Ans (2)

$$\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{400} = \frac{1}{4}$$

$$\therefore \eta = \frac{\text{Work done by engine}}{\text{Heat supplied by source}}$$

$$\frac{1}{4} = \frac{W}{40}$$

$$W = 10 \text{ kJ}$$

94. Ans (3)

AB temperature constant $V \downarrow$ so $P \uparrow$

in BC $T \propto V$, $P = \text{constant}$ $V \uparrow$

in CA volume constant $T \downarrow$ $P \downarrow$

95. Ans (2)

$$W = P_1 V_1 \ln \frac{V_2}{V_1}$$

$$W = 10 \times 10^3 \times 9 \times \ln \frac{27}{9}$$

96. Ans (3)

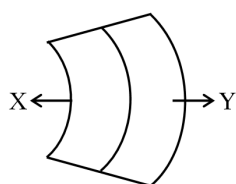
$\therefore$ For both spheres :

$$\frac{\Delta V}{V} = \gamma \Delta T \Rightarrow \frac{\Delta V}{V} = \frac{3\Delta d}{d} = \gamma \Delta T$$

Same for both spheres

97. Ans (2)

On cooling following happens



98. Ans (4)

During state change heat supplied does not increase temperature. So b to c & d to e are state change process.

99. Ans (3)

$$\therefore \frac{60 - 50}{10} = k(55 - 25) \dots (1)$$

$$\frac{50 - T}{10} = k \left(\frac{50 + T}{2} - 25 \right) \dots (2)$$

On solving - $T = 42.85^\circ\text{C}$

100. Ans (2)

In convection hot fluid goes upward due to low density and gravitation is necessary for it.

101. Ans (2)

$$Q = Q_r + Q_a + Q_t$$

$$400 = 120 + 120 + Q_t$$

$$Q_t = 160$$

$$= \frac{Q_t}{Q} \times 100\%$$

$$\frac{160}{400} \times 100\%$$

$$= 40\%$$

102. Ans (4)

$$\lambda_m \propto \frac{1}{T}$$

$$\frac{\lambda_{m1}}{\lambda_{m2}} = \frac{T_2}{T_1}$$

103. Ans (2)

$$E = \sigma AT^4$$

$$\frac{E_2}{E_1} = \left(\frac{T_2}{T_1} \right)^4$$

$$T_1 = 227 + 273 = 500 \text{ K}$$

$$\frac{E_2}{7} = \left(\frac{1000}{500} \right)^4 = 16$$

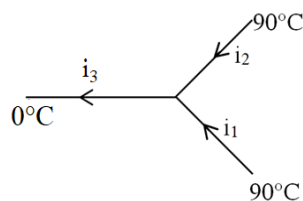
$$T_2 = 727 + 273 = 1000 \text{ K}$$

$$E_2 = 112 \text{ cal/cm}^2 \text{ sec.}$$

104. Ans (4)

At steady state different parts of conductor are at different temperatures but temperature of each part remains constant with time.

105. Ans (2)



$$\therefore i_3 = i_1 + i_2$$

$$\frac{T - 0}{R} = \frac{90 - T}{R} + \frac{90 - T}{R}$$

$$3T = 180^\circ$$

$$T = 60^\circ\text{C}$$

106. Ans (1)

$$n = \frac{1}{T} = \frac{\sqrt{k}}{2\pi\sqrt{m}} \quad n \propto \sqrt{\frac{k}{m}}$$

$$\frac{n'}{n} = \sqrt{\frac{k'}{m'}} \times \sqrt{\frac{m}{k}} = \sqrt{\frac{2k}{2m}} \times \sqrt{\frac{m}{k}} = 1$$

$$n = n$$

107. Ans (3)

$$\frac{1}{k_{eq}} = \frac{1}{k} + \frac{1}{2k} + \frac{1}{4k} + \frac{1}{8k} + \dots$$

$$\frac{1}{k_{eq}} = \frac{1}{k} \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right)$$

$$\frac{1}{k_{eq}} = \frac{1}{k} \left(\frac{1}{1 - 1/2} \right) = \frac{2}{k}$$

$$k_{eq} = \frac{k}{2}$$

108. Ans (1)

$$y_1 = 4 \sin(4\pi t + \pi/2) = 4\cos 4\pi t$$

$$y_2 = 3 \cos 4\pi t$$

Phase difference between both is zero, both are along the same line

$$A_{res.} = 4 + 3 = 7 \text{ m}$$

109. Ans (1)

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\pi} = 2\pi\sqrt{\frac{\ell}{g}}$$

$$\frac{\ell}{g} = \frac{1}{\pi^2}$$

$$\ell = 1 \text{ m}$$

110. Ans (2)

$$P.E = \frac{T \cdot E}{2}$$

$$\frac{1}{2} kx^2 = \frac{1}{2} ka^2$$

$$x = \frac{a}{\sqrt{2}} = \frac{6}{\sqrt{2}} = 3\sqrt{2} = 4.2 \text{ cm}$$

111. Ans (4)

$$T = 2\pi\sqrt{\frac{\ell}{g_{eff}}}$$

$$\therefore g_{eff} = 0$$

$$\therefore T = \infty$$

112. Ans (3)

$$T \propto \sqrt{\ell}$$

$$\frac{T'}{T} = \sqrt{\frac{16\ell}{\ell}} = 4$$

$$T' = 4T = 4 \times 2$$

$$T' = 8 \text{ sec}$$

113. Ans (1)

$$\frac{1}{2} m \omega^2 a^2 = 16$$

$$\omega^2 = \frac{32}{5.12 \times 625 \times 10^{-4}} ; \omega = 10$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{10} = \frac{\pi}{5}$$

114. Ans (4)

$$a = 2 \times 10^{-2} \text{ m}$$

$$V_{mean} = 12 \times 10^{-2} \text{ m/s}$$

$$\therefore V_{mean} = a\omega$$

$$\therefore 12 \times 10^{-2} = 2 \times 10^{-2} \times 2\pi n$$

$$\therefore n = \frac{3}{\pi} \text{ Hz}$$

115. Ans (2)

Acceleration in S.H.M is directly proportional to the displacement and opposite to the displacement.

116. Ans (3)

Initial and final position are same.

117. Ans (2)

$$x = A \sin \omega t$$

$$V = \omega A \cos \omega t = \omega A \sin(\omega t + \pi/2)$$

$$a = -\omega^2 A \sin \omega t = \omega^2 A \sin(\omega t + \pi)$$

118. Ans (1)

$$T = 2\pi\sqrt{\frac{\ell}{g_{eff}}} \quad g_{eff} = g + a \text{ (when lift accelerates upwards)}$$

119. Ans (1)

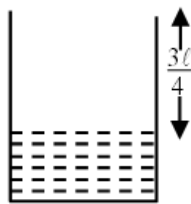
Standard equation $y = 2a \cos kx \sin \omega t$

By comparing this equation with given equation

$$K = \frac{2\pi}{\lambda} = \frac{\pi}{20} \Rightarrow \lambda = 40 \text{ cm}$$

$$\text{Distance between two nodes} = \frac{\lambda}{2} = \frac{40}{2} = 20 \text{ cm}$$

120. Ans (3)



$$\frac{V}{4l} = 220 \text{ Hz}$$

$$n' = \frac{3}{4} \frac{V}{\frac{3}{4}l} = \frac{V}{l} = 880 \text{ Hz}$$

121. Ans (4)

$$I = 2\pi^2 n^2 a^2 \rho V$$

$$\frac{I_1}{I_2} = \frac{n_1^2}{n_2^2} \times \frac{a_1^2}{a_2^2} = \frac{1}{4} \times \frac{36}{9} = 1$$

122. Ans (3)

Second overtone means three loops :

$$\therefore 3 \frac{\lambda}{2} = L \text{ or } \lambda = \frac{2L}{3}$$

Anti-nodes will be obtained at $\frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}$ or

$$\frac{L}{6}, \frac{3L}{6}, \frac{5L}{6}$$

123. Ans (1)

$$n = \frac{P}{2l} \sqrt{\frac{T}{\pi r^2 \rho}} \quad n \propto \frac{1}{\sqrt{\rho}}$$

Graph between n and $\sqrt{\rho}$ will be hyperbola.

124. Ans (2)

$$\begin{aligned} \text{Phase difference } \Delta\phi &= \frac{2\pi}{\lambda} \Delta x = K \Delta x \\ &= \pi \times 0.01 \times 25 = \frac{\pi}{4} \end{aligned}$$

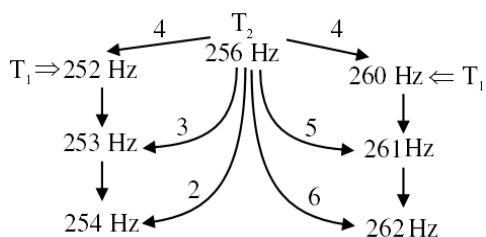
125. Ans (1)

$$V = \frac{\omega}{K} = \frac{120\pi}{\frac{2\pi}{5 \times 10^{-2}}} = 3 \text{ m/s}$$

$$V = \sqrt{\frac{T}{m}} \Rightarrow 3 = \sqrt{\frac{T}{0.4}} \Rightarrow T = 9 \times 0.4$$

$$T = 3.6 \text{ N}$$

126. Ans (1)



The beat frequency should increase, so answer is 260 Hz.

127. Ans (2)

Sound waves requires medium for propagation.

128. Ans (1)

$$V \propto \lambda$$

depends on medium

$$V = f\lambda$$

f not depends on medium, it depends on source.

129. Ans (2)

$$I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$

$$I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$

$$I_{\max} + I_{\min} = 2(I_1 + I_2)$$

130. Ans (2)

$$V_P = 3V_w$$

$$a_0 \omega = 3n\lambda$$

$$a_0 \times 2\pi n = 3n\lambda$$

$$\lambda = \frac{2\pi a_0}{3}$$

131. Ans (3)

NCERT Pg. # 284

$$\frac{V_P}{V_w} = \frac{\omega A}{\omega/K} = AK = 60 \times 6 = 360$$

132. Ans (3)

Frequency of first and last tuning fork are $2n$ & n respectively.

$$2n - 24 \times 3 = n \Rightarrow n = 72 \text{ Hz}$$

Frequency of 21st tuning fork = $(2 \times 72 - 20 \times 3)$

$$= 84 \text{ Hz}$$

133. Ans (2)

$$\frac{V}{(40 + 2e)} = \frac{3V}{4(31 + e)}$$

$$3(40 + 2e) = 4(31 + e)$$

$$2e = 4$$

$$e = 2 \text{ cm}$$

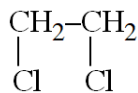
134. Ans (2)

Amplitude and energy are exponential function of time.

135. Ans (4)

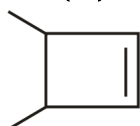
Resonance frequency is independent from the density of reflecting medium (denser medium)

136. Ans (1)



ethylene chloride (no π bonds)

141. Ans (3)



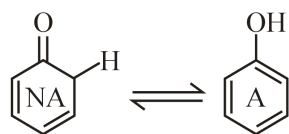
Can show GI along ring (Restricted rotation along σ bond)

142. Ans (1)

Fact

143. Ans (1)

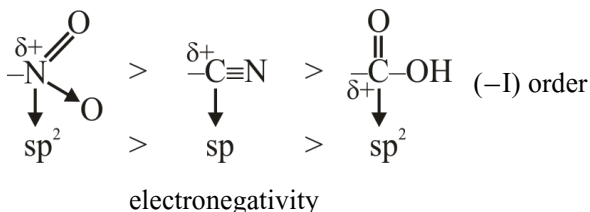
Enol form is aromatic in nature



145. Ans (1)

According to pK_a value

146. Ans (4)



148. Ans (2)



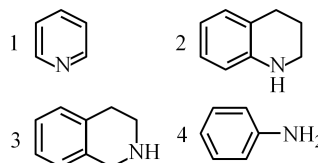
149. Ans (3)

Heat of hydrogenation $\propto \frac{1}{\text{stability}}$

151. Ans (3)

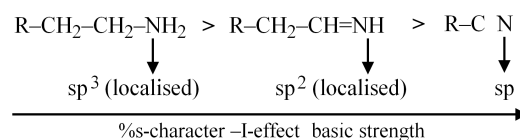
A. Question: The question asks which of the following is most basic

B. Given Data: Options are:

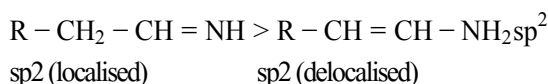


C. Concept:

- Basic strength of nitrogenous compound depends upon the hybridised state of nitrogen.

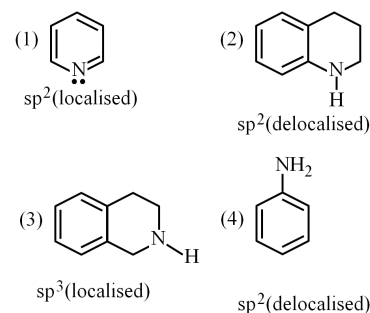


- If %s-character is same then $K_b = \text{localised lone pair} > \text{delocalised lone pair}$



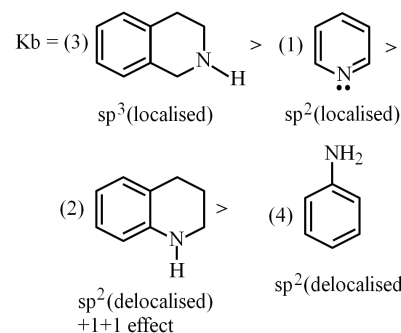
The -NH₂ group is attached directly to the conjugated system or benzene ring. It results in the unshared electron pair on nitrogen atom to be in conjugation with the conjugated system or benzene ring and thus making it less available for protonation and decreases basic strength.

D. Explanation:



$K_b = \text{sp}^3 > \text{sp}^2 \text{ (localised lone pair)} > \text{sp}^2 \text{ (delocalised lone pair)}$

$K_b = (3) > (1) > (2), (4)$



E. Final Answer: 3

154. Ans (1)

Fact based

155. Ans (1)

Compounds purified by steam distillation which are immiscible in water but steam volatile.

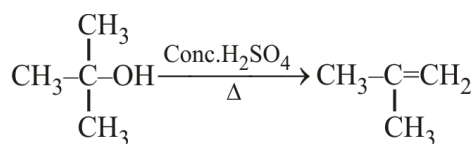
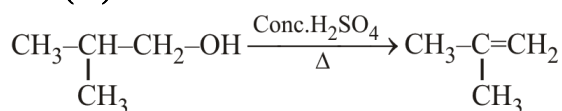
157. Ans (2)

$$\%N = \frac{1.4}{w} \times NV = \frac{1.4}{0.1} \times 4 = 56\%$$

159. Ans (3)

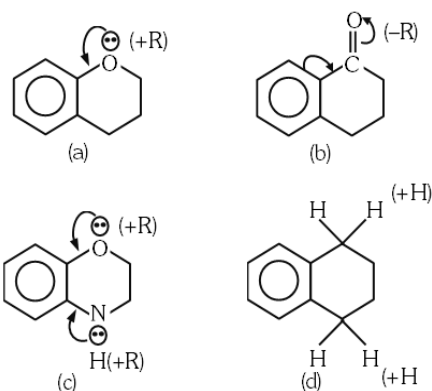
Reaction in column I are oxidation of alkane under different condn.

161. Ans (4)



163. Ans (2)

Reactivity towards electrophilic aromatic substitution reaction (ESR) $\propto$ density in ring.

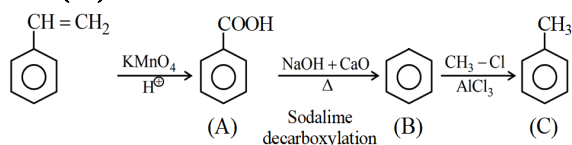


(2) $c > a > d > b$

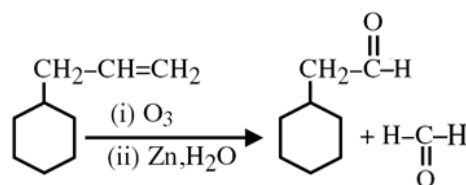
164. Ans (1)

Hydro boration oxidation

165. Ans (2)



166. Ans (1)



167. Ans (1)

Na/Liquid NH_3 gives trans-alkene with nonterminal alkynes.

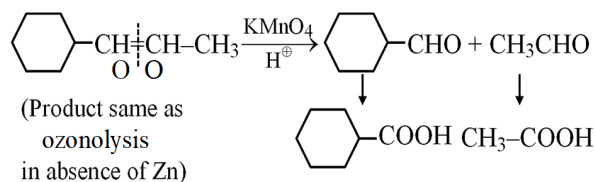
169. Ans (4)

Rate of Elimination $\propto$ Stability of alkene.

171. Ans (3)

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173. Ans (2)



174. Ans (3)

Aniline do not show FCR.

175. Ans (3)

A = Benzene B = Friedel craft reaction

176. Ans (3)

Silver acetylides are obtained by passing 1-alkyne in the ammonical solution of silver nitrate (tollen's reagent)

179. Ans (1)

Fact Based